

Decentralized Swarm Resilience Under Stochastic Gamma Noise: A Computational Model for Pressure Vessel Inspection

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Abstract—Robotic PWR inspection combines turbulent flow with spatially heterogeneous gamma radiation, degrading motion, sensing, and communication. This paper frames inspection as a hardware-fault-tolerant control problem and presents a decentralized, communication-degradation-tolerant Modified Particle Swarm Optimization (M-PSO) framework for micro-agent navigation. A reduced-order simulator couples hydrodynamic drag, a non-homogeneous dose field, dose-dependent sensor noise and packet loss, and permanent cumulative-dose latch-up. M-PSO propagates neighborhood-local information through a per-agent channel state rather than a privileged global leader. In 500-trial Monte Carlo campaigns, decentralized M-PSO achieves 84.0% target-localization success (95% CI [80.5, 87.0]) under the modeled Extreme profile, versus 3.0% for a static-leader centralized baseline and 1.8% for a re-election variant. A separate hardened comparison sweeps a state-replicating hot-standby timeout of $h = 1, 3, 5$ missed heartbeats: success remains 0.6–1.6% although failover activates in 78.6–81.2% of trials. New sensitivity studies select $N = 25$ as a coverage-dose knee and show decentralized performance remains 83.0–86.0% across $\Lambda_{\text{fail}} = 100\text{--}800$ Gy. The findings are limited to the stated computational model but support decentralized coordination when hazard-driven communication degradation is explicit.

Index Terms—Harsh-environment robotics, resilient path planning, sensor-noise robustification, swarm robotics, particle swarm optimization, decentralized control, nuclear inspection, fault-tolerant communication.

I. INTRODUCTION

The primary coolant loop of a Pressurized Water Reactor (PWR) is a physically inaccessible, hazardous environment subject to mandatory in-service inspection, combining fully turbulent flow ($Re > 10^5$) with intense, non-homogeneous gamma radiation capable of degrading unshielded microelectronics on contact. No mature, field-proven autonomous solution has been reported for this specific dual-hazard interior [1]. Related field systems include the MALLARD autonomous surface vehicle for wet nuclear storage and a radiological-characterization ROV [2], [3], but neither addresses decentralized swarm navigation in a turbulent, radiation-degrading PWR channel.

Existing technologies are ill-suited to it. Tethered Remotely Operated Vehicles (ROVs), the incumbent underwater-inspection solution, suffer tether snagging and fatigue frac-

ture under turbulent shear; fiber-optic tethers substituted to avoid EMI instead suffer radiation-induced attenuation (RIA), progressively severing the channel [4]. The static-leader architecture examined here has a designated agent as a single point of failure: dose accumulation, latch-up, or single-event upset can terminate its estimate propagation. This concern is consistent with field-documented fragility of teleoperated robotics in high-radiation environments [5]; recent multi-robot nuclear-decommissioning demonstrations also retain a human-in-the-loop coordinator as an analogous aggregation point [6].

This motivates decentralized, bio-inspired coordination, a direction consistent with the broader swarm-robotics literature’s continued shift toward decentralization [7]. Particle Swarm Optimization (PSO) [8] is attractive precisely because no single agent’s failure is catastrophic to the collective search. Each agent’s canonical velocity update is

$$v_i^{t+1} = \omega v_i^t + c_1 r_1 (p_i^{\text{best}} - x_i^t) + c_2 r_2 (g^{\text{best}} - x_i^t), \quad (1)$$

with $r_1, r_2 \sim U(0, 1)$ and the standard position update $x_i^{t+1} = x_i^t + v_i^{t+1}$. This presupposes flawless, zero-latency transmission of p_i^{best} and g^{best} —an assumption categorically violated in a radiologically active, turbulent PWR loop, where both physical and communicated agent state are subject to continuous stochastic corruption.

Scoping the contribution. Neighborhood-restricted social attraction is itself an established PSO variant [9], [10]; it is not introduced here. Nor is it a new theoretical result that a no-single-point-of-failure topology degrades gracefully while leader-follower coordination collapses—this instances known properties from distributed consensus [11]. The contribution here is narrower: a *physically-grounded process for generating the neighborhood topology itself*. In this process, graph edges are jointly gated by inter-agent range and by a radiologically-driven, dual-mechanism hardware-failure model (transient Bernoulli drops and permanent cumulative-dose latch-up), and are further coupled to per-agent hydrodynamic drag specific to a turbulent PWR channel. The closest prior work, Ristic et al. [12], addresses estimation-theoretic source localization under turbulent dispersion; this paper instead addresses swarm-optimization kinematics in which the *communication medium*

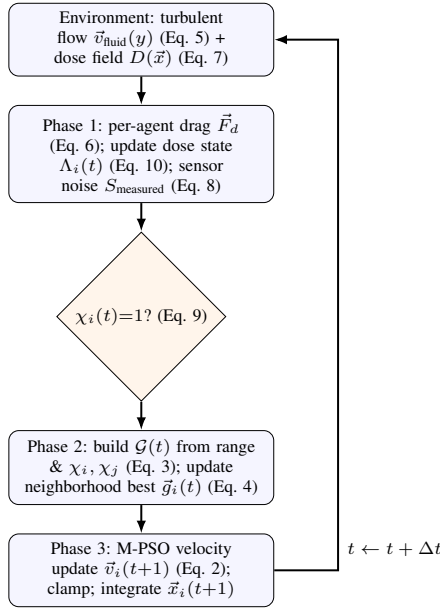


Fig. 1. Per-timestep system workflow. A failed channel removes social information but does not halt the physical update.

itself is degraded by the hazard. Finally, a reduced-order, purpose-built simulator is used in place of a general-purpose platform [13], since this keeps a $500 \times 3 \times 3$ Monte Carlo campaign computationally tractable—a pragmatic trade-off, not a fidelity claim (Sec. IV).

The contributions are threefold:

- 1) **Hydrodynamic Agent Kinematics:** per-agent quadratic drag, $\vec{F}_d = -\frac{1}{2}\rho C_d A |\vec{v}_{\text{rel}}| \vec{v}_{\text{rel}}$ ($\vec{v}_{\text{rel}}$: velocity relative to local fluid), coupled to PSO kinematics under $Re > 10^5$ flow, replacing idealized frictionless motion.
- 2) **Non-Homogeneous Radiation-Induced Degradation:** a spatially heterogeneous gamma dose field driving dose-proportional sensor noise and dose-dependent communication loss (Sec. II-C).
- 3) **Decentralized Adaptive Swarm Mechanisms:** neighborhood-restricted information exchange gated by a per-agent, radiologically-driven channel-state variable, preserving search efficiency under dropout in the evaluated leader-based baselines.

II. SYSTEM ARCHITECTURE AND MATHEMATICAL MODEL

Three subsystems integrate at each timestep: M-PSO kinematics and communication, hydrodynamic drag, and dual-phenomenon radiation degradation. Fig. 1 summarizes the workflow.

A. Kinematic Swarm Update System

The motion of agent $i \in \{1, \dots, N\}$ is governed by an M-PSO update augmenting the canonical cognitive-social formulation with a physical forcing term from fluid-structure interaction:

$$\begin{aligned} \vec{v}_i(t+1) = & w \vec{v}_i(t) + c_1 r_1 (\vec{p}_i(t) - \vec{x}_i(t)) \\ & + \chi_i(t) c_2 r_2 (\vec{g}_i(t) - \vec{x}_i(t)) + \frac{\vec{F}_d}{m} \Delta t, \quad (2) \end{aligned}$$

with forward-Euler position update $\vec{x}_i(t+1) = \vec{x}_i(t) + \vec{v}_i(t+1) \Delta t$. The term $(\vec{F}_d/m) \Delta t$ is the principal departure from canonical PSO, embedding Newtonian drag directly into search kinematics. Following PSO convention [8], [14], the cognitive/social terms are an abstract velocity-space input, not a physical force; drag, by contrast, is a genuine kinematic acceleration, explicitly integrated over Δt to combine correctly with the algorithmic terms rather than being added as a raw acceleration—precluding agents from executing arbitrarily precise trajectories in a medium that physically resists motion. Eq. (2) also replaces the canonical global attractor $\vec{g}^{\text{best}}$ with an agent-specific, neighborhood-local attractor $\vec{g}_i(t)$, defined next.

1) *Dynamic Communication Graph:* Let $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t))$ denote the swarm communication graph, $\mathcal{V} = \{1, \dots, N\}$. An edge $(j, i) \in \mathcal{E}(t)$ —a successful transmission from j to i , over a transceiver of fixed physical range r_{comm} —requires joint range and hardware operability:

$$\|\vec{x}_j(t) - \vec{x}_i(t)\| \leq r_{\text{comm}} \text{ and } \chi_j(t) = 1 \text{ and } \chi_i(t) = 1, \quad (3)$$

where $\chi_k(t) \sim \text{Bernoulli}(1 - P_{\text{drop}}(\vec{x}_k(t)))$ is the per-agent channel state (Sec. II-C), used for both transmit and receive. Thus $\mathcal{N}_i(t)$ collapses to $\{i\}$ when $\chi_i(t) = 0$; the explicit social-term gate is a defensive implementation choice, not a separate failure model.

2) *Neighborhood-Restricted Local Best:* The active neighborhood of agent i is $\mathcal{N}_i(t) = \{j \in \mathcal{V} \mid (j, i) \in \mathcal{E}(t)\} \cup \{i\}$, giving a decentralized, neighborhood-specific attractor

$$\vec{g}_i(t) = \arg \max_{j \in \mathcal{N}_i(t)} \{f(\vec{p}_j(t))\}, \quad (4)$$

where $f(\cdot)$ is the noise-corrupted signal fitness. Because $\mathcal{N}_i(t)$ is a stochastic function of $\mathcal{G}(t)$, $\vec{g}_i(t)$ may differ across agents and, in the limit, reduce to full isolation ($\mathcal{N}_i(t) = \{i\}$) when all inbound edges fail simultaneously.

3) *Notation:* $\vec{x}_i, \vec{v}_i \in \mathbb{R}^2$ are position/velocity; $\vec{p}_i$ and $\vec{g}_i$ are personal and neighborhood-local best states under a maximization fitness f ; w, c_1, c_2 are fixed PSO weights; and $\vec{F}_d/m$ is drag acceleration. Packet loss is represented only by $\chi_i(t)$, not by changing c_2 .

B. Turbulent Hydrodynamics and Drag Modeling

Unlike the inert medium assumed by idealized PSO, the PWR coolant pipe presents a fully turbulent flow field that continuously perturbs agent motion. For fully developed flow at Reynolds number $Re > 10^5$, the cross-sectional velocity is well-approximated by the empirical one-seventh power-law profile [15] plus a stochastic fluctuation term:

$$\vec{v}_{\text{fluid}}(y) = v_{\text{max}} \left(1 - \frac{|y|}{R}\right)^{\frac{1}{7}} \hat{i} + \vec{v}_{\text{turb}}, \quad \vec{v}_{\text{turb}} \sim \mathcal{N}(0, \sigma_{\text{turb}}^2), \quad (5)$$

where $y \in [-R, R]$ is the wall-normal coordinate, R the pipe radius, v_{max} the centerline velocity, and $\hat{i}$ the pipe's axial unit vector; the profile enforces no-slip at $y = \pm R$ and a maximum at $y = 0$. $\vec{v}_{\text{turb}}$ is a first-order stochastic representation of the Reynolds-decomposed fluctuation $u' = u - \bar{u}$, omitting full

RANS closure for tractability while preserving the essential stochastic forcing.

This field couples to agent motion via the standard quadratic drag relation for bluff bodies:

$$\vec{F}_d = -\frac{1}{2}\rho C_d A \|\vec{v}_i(t) - \vec{v}_{\text{fluid}}(y)\|^2 \hat{u}_d, \quad (6)$$

with $\rho \approx 1000 \text{ kg/m}^3$ coolant density, C_d the drag coefficient, A the frontal area, and $\hat{u}_d$ the unit vector opposing relative velocity. This quadratic dependence is strongly non-linear: agents moving against the mean flow near the steep-gradient wall region experience disproportionate resistive forcing, biasing trajectories toward the bulk flow direction—a bias absent from idealized PSO.

C. Ionizing Radiation Degradation Fields

A spatially non-homogeneous gamma field, concentrated near the fissure (corrosion/activation sites co-locate in practice), is modeled as a background term plus inverse-square enhancement:

$$D(\vec{x}) = D_{\text{bg}} + \frac{K}{\|\vec{x} - \vec{x}_f\|^2 + 1.0}, \quad (7)$$

with D_{bg} ambient dose rate, K source intensity, and the $+1.0$ regularizer preventing a singularity at $\vec{x} \rightarrow \vec{x}_f$ while permitting severe local dose—an unavoidable mission-success/survivability trade-off, since inspection requires approaching the highest-hazard region.

This drives two mechanistically distinct degradation phenomena [16]. **(a) Sensor noise:**

$$S_{\text{measured}}(\vec{x}) = S(\vec{x}) + \epsilon_\gamma, \quad \epsilon_\gamma \sim \mathcal{N}(0, \sigma_\gamma^2 D(\vec{x})), \quad (8)$$

where $S(\vec{x})$ is the true, noise-free proximity signal used as $f(\cdot)$ throughout (peaked at $\vec{x}_f$, decaying with distance), and noise variance scales linearly with dose rate via σ_γ^2 , so reliability degrades as agents approach the region of interest.

(b) Communication loss:

$$P_{\text{drop}}(\vec{x}) = 1 - e^{-\beta D(\vec{x})}, \quad (9)$$

with β the hardware's dose-to-failure sensitivity ($P_{\text{drop}} \rightarrow 1$ at high dose; $P_{\text{drop}} \rightarrow \beta D(\vec{x})$ at low dose, consistent with a Poisson-thinned first-order failure process). A drop instantaneously zeroes $\chi_i(t)$ (Eq. 3), collapsing Eq. (2) to purely cognitive-inertial motion for that agent/step—informational isolation, not agent failure, since connectivity may be stochastically restored.

This transient mechanism alone cannot capture *permanent* hardware failure under sustained exposure. Each agent therefore accumulates a running dose state

$$\Lambda_i(t) = \Lambda_i(t-1) + D(\vec{x}_i(t)) \Delta t, \quad \Lambda_i(0) = 0, \quad (10)$$

and once $\Lambda_i(t) \geq \Lambda_{\text{fail}}$, the transceiver latches up permanently:

$$\Lambda_i(t) \geq \Lambda_{\text{fail}} \implies \chi_i(t') = 0 \quad \forall t' \geq t, \quad (11)$$

regardless of subsequent position, where Λ_{fail} is a fixed, hardware-specific cumulative-dose failure threshold (Table I). Only Eq. (11) is permanent; ordinary Bernoulli drops via $P_{\text{drop}}(\vec{x})$ alone are always transient. It is the swarm's aggregate resilience under this combination of transient isolation and occasional permanent hardware loss—not any individual agent's performance—that is investigated below.

TABLE I
SIMULATION CONFIGURATION. PROFILES USE
(D_{BG}, K) = (0, 0), (0.5, 10), (2, 100).

Parameter	Symbol	Value
Pipe geometry	L, R	10.0, 1.0 m
Fissure / agent mass	$\vec{x}_f, m$	(8.0, 0.9) m, 0.05 kg
Flow / turbulence	$v_{\text{max}}, \sigma_{\text{turb}}$	1.5, 0.1 m/s
Drag / density	C_d, A, ρ	0.47, $1.26 \times 10^{-3} \text{ m}^2$, 1000 kg/m ³
Swarm / horizon	N, t_{max}	25, 150
Step / range / clamp	$\Delta t, r_{\text{comm}}, v_{\text{clamp}}$	0.1 s, 2.0 m, 3.0 m/s
PSO weights	w, c_1, c_2	0.6, 1.2, 1.2
Channel / latch-up	$\beta, \sigma_\gamma, \Lambda_{\text{fail}}$	0.15 s/Gy, 0.08, 300 Gy

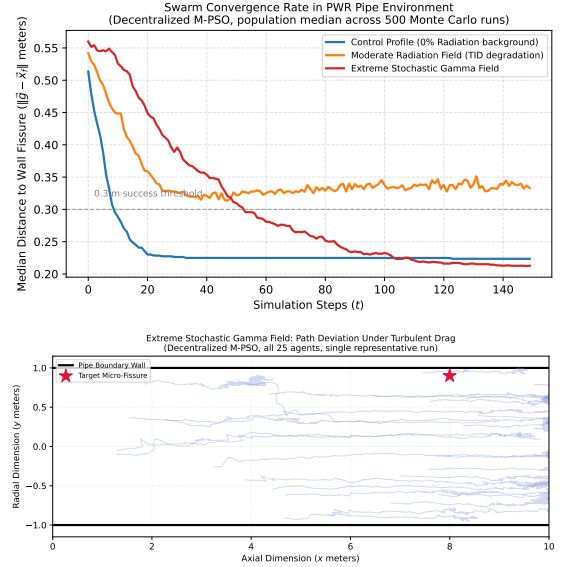


Fig. 2. Decentralized M-PSO behavior: (top) population-median convergence distance across 500 runs per profile; (bottom) a representative Extreme-profile trajectory realization (25 agents).

Initial positions are drawn uniformly from $\Omega_{\text{pipe}} = [0, L] \times [-R, R]$; a fixed v_{clamp} is applied identically to all architectures. The per-timestep phases are the physical/radiological update, graph construction and local-best update, then velocity/position integration (Fig. 1).

III. SIMULATION CONFIGURATION AND RESULTS ANALYSIS

The simulator executes $N = 25$ agents for 150 steps in a 2D pipe cross-section, with a near-wall fissure at (8.0, 0.9) m. Control, Moderate, and Extreme profiles use the three dose-field settings in Table I. A run succeeds if its best estimate comes within 0.3 m of $\vec{x}_f$. All results are generated by `pso_nuclear_sim.py` as independent SHA-256-seeded 500-trial campaigns; velocity clamping is identical across architectures.

Choice of N and t_{max} . The 20 m² domain is swept over $N = 5$ –50 (Table IV); success rises steeply through $N = 25$ while cumulative dose and $O(N^2)$ graph cost continue to grow. Thus $N = 25$ is an empirical coverage-dose knee, not an a priori choice. $t_{\text{max}} = 150$ (15 s) exceeds the observed Control/Moderate plateau and captures the slower Extreme tail.

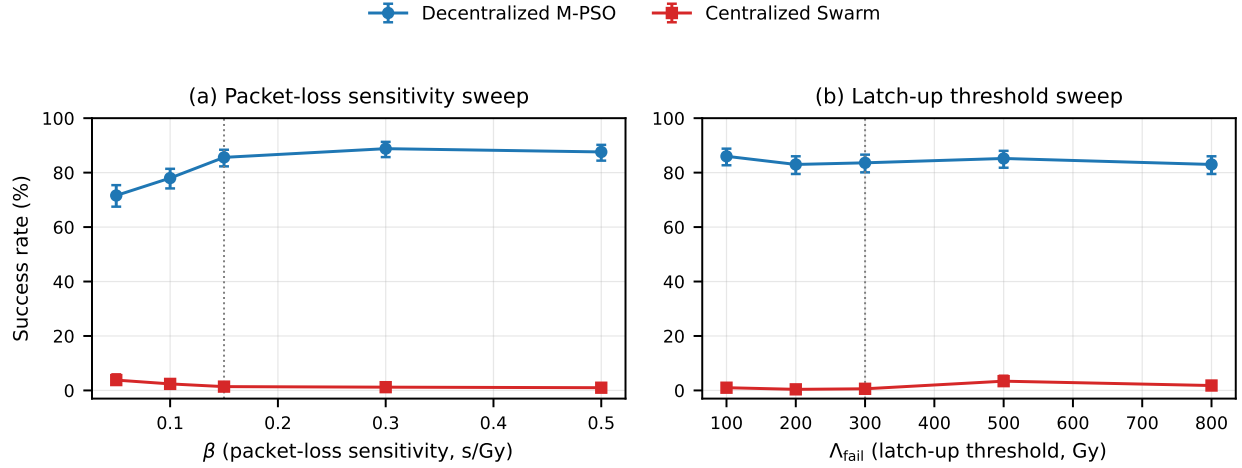


Fig. 3. Full communication-fragility campaign (Extreme profile; 500 independent trials per point). Markers show success rate and bars show Wilson 95% confidence intervals; dotted lines mark the nominal settings. Decentralized M-PSO remains robust to the Λ_{fail} sweep, whereas a static leader remains unsuccessful even when hardening greatly reduces leader latch-up.

A. Baseline Comparison

A Monte Carlo campaign (500 runs/architecture/profile) compares Decentralized M-PSO against two centralized variants and a Canonical PSO. **Centralized Swarm (static leader):** a single leader broadcasts $\vec{g}^{\text{best}}$, freezing permanently after leader latch-up ($\chi_{\text{leader}}(t) = 0$). **Centralized Swarm (re-election):** an inoperative leader is replaced by the highest-fitness operational, non-latched agent. A post-review hardening check adds one **hot standby**: the nearest non-leader replicates state only when both radios and range permit, then promotes after $h \in \{1, 3, 5\}$ consecutive missed heartbeats (timeouts fixed before inspection). These baselines isolate leader dependence through one redundancy level; multi-standby and quorum protocols remain outside scope. **Canonical PSO:** fully-connected, no drag, no communication-degradation model. Performance is Success Rate (fraction with $\|\vec{g} - \vec{x}_f\| \leq 0.3$ m, with n_{success} and Wilson-score 95% CI), Convergence Time, and Cumulative Swarm Dose (Eq. 10). Rates also control for trivial spawns: with positions drawn over $\Omega_{\text{pipe}} (\approx 20 \text{ m}^2)$, an agent can occasionally start within the success radius by chance; non-trivial (search-driven) rates excluding such runs are reported directly in Table II and discussed below.

Control and Moderate. In Control, all architectures have overlapping CIs (57.0–66.0%); residual failures are consistent with PSO stagnation once personal and neighborhood best states coincide [17]. Moderate radiation reduces decentralized success to 58.0% while the static leader falls to 29.8%. Re-election partially recovers to 40.6% because communication remains sufficiently available for the election step.

Extreme. Decentralized M-PSO reaches 84.0% [80.5, 87.0], versus 3.0% and 1.8% [0.9, 3.4] for the static and re-election baselines. Canonical PSO reaches 29.6% but omits the modeled communication hazard. In the independent hardening campaign, decentralized success is 87.0% [83.8, 89.7], while

all hot-standby variants remain near zero (Table VI). Failover completes in 78.6–81.2% of trials, so the low success is not an inert-failover artifact. Promotion changes the coordinator, not the information path: reporting and replication use the same range/ χ graph, and both coordinator agents accrue dose. Within the evaluated architectures, one backup therefore does not restore edges lost under Extreme fragmentation.

Ablation isolation. Table III isolates drag-awareness, the explicit $\chi_i(t)$ social-term gate, and the two radiological channels, across all seven configurations (Extreme profile). Removing drag alone (cell B) collapses Decentralized M-PSO to 23.8%, with *zero* non-trivial success—without drag compensation, agents cannot meaningfully navigate the turbulent flow. Adding drag to the non-decentralized Canonical PSO architecture (cell D) yields 38.0% [33.9, 42.3]^d—a modest +8.4-point improvement over the drag-free Canonical baseline (29.6%, Table II)—while disabling communication degradation entirely (cell C) reaches 44.8%. Drag therefore has a real but modest architecture-independent effect, and a much larger effect with decentralized topology (A vs. B: +60.6 points). The corresponding difference-in-differences interaction is 52.2 points (Wald $z = 13.4$, $p < 10^{-40}$; 95% CI [44.6, 59.8] points), supporting an interaction rather than a decentralization-exclusive drag effect. The explicit $\chi_i(t)$ social-term gate is not load-bearing: cell E gives 83.8% [80.3, 86.8], within noise of cell A’s 84.4%. The radiological channels are non-additive: sensor noise alone is harmful (41.8%, F), while packet-loss only gives 93.2% (466/500, G) versus 84.4% (422/500, A), an 8.8-point difference (two-sided two-proportion $z = 4.4$, $p < 10^{-4}$; 95% difference CI [4.9, 12.7] points). In this reduced-order model, fragmentation without noisy fitness signals behaves as a diversity mechanism; it is not a recommendation to increase packet loss in deployed systems.

Taken together, these results show a large architecture-

TABLE II
ARCHITECTURE COMPARISON (500 INDEPENDENT TRIALS PER PROFILE/ARCHITECTURE). VALUES ARE SUCCESS [WILSON 95% CI], NON-TRIVIAL (SEARCH-DRIVEN) SUCCESS, AND MEAN CUMULATIVE DOSE.

Profile	Architecture	Success (%) [95% CI]	n_{success}	Non-triv. (%)	Cum. Dose (Gy)
Control	Decentralized M-PSO	66.0 [61.7,70.0]	330/500	55.6	0.0
	Centralized (static leader)	62.0 [57.7,66.1]	310/500	50.4	0.0
	Centralized (re-election)	60.2 [55.8,64.4]	301/500	49.5	0.0
	Canonical PSO	57.0 [52.6,61.3]	285/500	44.3	0.0
Moderate	Decentralized M-PSO	58.0 [53.6,62.2]	290/500	46.3	1146.9
	Centralized (static leader)	29.8 [26.0,34.0]	149/500	20.5	1387.6
	Centralized (re-election)	40.6 [36.4,45.0]	203/500	31.5	1427.6
	Canonical PSO	42.4 [38.1,46.8]	212/500	24.1	3002.4
Extreme	Decentralized M-PSO	84.0 [80.5,87.0]	420/500	79.9	9490.6
	Centralized (static leader)	3.0 [1.8,4.9]	15/500	0.8	12610.4
	Centralized (re-election)	1.8 [0.9,3.4]	9/500	0.8	10328.1
	Canonical PSO	29.6 [25.8,33.7]	148/500	12.9	25470.6

Each table row is independently SHA-256-seeded; nominally identical settings are independent replications, not shared draws. Four nominal decentralized campaigns span 83.4–87.0%, with overlapping Wilson CIs, providing an internal stability check.

TABLE III
FULL ABLATION ISOLATION (EXTREME PROFILE, $N = 25$, $r_{\text{comm}} = 2.0$ m; 500-TRIAL MONTE CARLO CAMPAIGN PER CELL, SHA-256-SEEDED PER CONDITION IDENTITY)

Configuration	Success (%) [95% CI]	Non-triv. (%)
A. Full M-PSO model (drag, explicit χ -gate, both degradation channels)	84.4 [81.0, 87.3]	79.9
B. No drag term	23.8 [20.3, 27.7]	0.0
C. No communication degradation ($\chi_i(t) \equiv 1$; fully connected, no latch-up)	44.8 [40.5, 49.2]	30.9
D. Canonical PSO + drag (non-decentralized)	38.0 [33.9, 42.3]	26.4
E. No explicit χ -gate (self-collapse of $\mathcal{N}_i(t)$ only)	83.8 [80.3, 86.8]	79.0
F. Sensor-noise only (no packet loss; latch-up still active)	41.8 [37.6, 46.2]	28.0
G. Packet-loss only (no sensor noise)	93.2 [90.6, 95.1]	91.3

Cell A replicates the nominal Table II configuration independently (see footnote a there). Cells E vs. A isolate the explicit $\chi_i(t)$ social-term gate: 83.8% vs. 84.4%, CIs almost fully overlapping—the gate offers only a marginal benefit over neighborhood self-collapse alone (Sec. II-A1). Cell D vs. the Canonical PSO row of Table II (29.6%) isolates drag’s main effect independent of topology: 38.0% [33.9, 42.3] is a modest but real +8.4-point improvement over the drag-free canonical baseline (CIs only marginally overlapping, at 33.7–33.9%). This is far smaller than drag’s effect within the decentralized architecture (cf. A vs. B, 84.4% vs. 23.8%, a +60.6-point gain), indicating a strong positive interaction between drag-awareness and decentralized topology rather than an architecture-independent main effect alone.^d

^dCorrected value. An earlier version of the simulator’s drag-application logic evaluated `architecture=="Canonical PSO"` or not `use_drag` as a single boolean gate, which unconditionally skipped drag for the Canonical PSO architecture regardless of the `use_drag` flag. This silently defeated cell D’s intended configuration (drag genuinely enabled on a non-decentralized architecture); the previously reported 26.6% [22.9, 30.6] reflected two independently-seeded *drag-free* Canonical PSO runs, not a drag-vs-no-drag comparison. The bug is fixed in the released simulator (Sec. IV) by making drag application depend solely on `use_drag`, independent of architecture; 38.0% above is the corrected, re-run result under the same $n=500$ trial count and condition identity.

dependent advantage under the modeled Extreme profile: relative to the evaluated static-leader and re-election baselines, decentralized M-PSO reaches 84.0% versus 3.0% and 1.8%. This comparison isolates the vulnerability of leader dependence under coupled radiation and communication degradation; it does not establish that every possible fault-tolerant centralized design would fail.

B. Sensitivity to Swarm Size and Communication Range

The results above fix $N = 25$ and $r_{\text{comm}} = 2.0$ m (Table I). To characterize robustness and generalization beyond the noise/radiation axis already varied across profiles, two further Monte Carlo sweeps (500 trials per condition, Extreme profile,

TABLE IV
SWARM-SIZE SENSITIVITY (EXTREME PROFILE; 500 INDEPENDENT TRIALS PER N).

N	Success (%)	95% CI	Mean Conv. (steps)	Cum. Dose (Gy)
10	59.4	[55.0, 63.6]	49.8	3717.1
25	83.4	[79.9, 86.4]	30.6	9489.6
50	98.8	[97.4, 99.4]	17.0	19208.9

TABLE V
COMMUNICATION-FRAGILITY SENSITIVITY (EXTREME PROFILE; 500 INDEPENDENT TRIALS PER ROW). SUCCESS IS [WILSON 95% CI]. D : DECENTRALIZED; C : STATIC LEADER.

Setting	D success	C success	C latch-up
$\beta = 0.05$ s/Gy	71.6 [67.5,75.4]	3.8 [2.4,5.9]	92.8%
$\beta = 0.50$ s/Gy	87.6 [84.4,90.2]	1.0 [0.4,2.3]	52.6%
$\Lambda_{\text{fail}} = 100$ Gy	86.0 [82.7,88.8]	1.0 [0.4,2.3]	100.0%
$\Lambda_{\text{fail}} = 800$ Gy	83.0 [79.5,86.0]	1.8 [0.9,3.4]	9.4%

identical success threshold) vary these system parameters directly.

Swarm size. Table IV shows the coverage–dose trade-off: performance rises steeply through $N = 25$, while dose grows linearly. Larger swarms improve success but duplicate coverage and increase exposure, motivating the selected knee.

Neighbor and channel failure dynamics. Eq. 3 applies to every transceiver, not only a leader. The original range sweep left decentralized success within 80.4–87.0% over $r_{\text{comm}} = 0.5$ –3.0 m, while the static leader remained 0.6–2.4%. Table V adds hardware/channel sensitivity: decentralized performance remains 83.0–86.0% across $\Lambda_{\text{fail}} = 100$ –800 Gy, whereas hardening the leader lowers its latch-up incidence without restoring centralized success. The β trend is a model-specific observation, not a recommendation to increase packet loss.

Relation to learning-based coordination. QMIX and MAPPO are relevant centralized-training/decentralized-execution baselines for cooperative control [18], [19]. They are not assigned numerical results here: a defensible comparison requires the same local observations, hazard process, training budget, and held-out degradation shifts. In contrast, the present M-PSO controller is training-free and exposes its dose and communication gates explicitly; Fig. 3 therefore establishes robustness only for this model, rather than claiming superiority over learned policies.

TABLE VI
SINGLE-HOT-STANDBY ROBUSTNESS (EXTREME; 500 TRIALS/h).
SUCCESS: [WILSON 95% CI]; NON-TRIV.: % (SUCCESSES/ELIGIBLE
STARTS).

h	Success (%) [95% CI]	Failover (%)	Non-triv. (%)
1	1.4 [0.7,2.9]	81.2	0.26 (1/391)
3	1.6 [0.8,3.1]	80.4	0.00 (0/392)
5	0.6 [0.2,1.7]	78.6	0.00 (0/393)

IV. CONCLUSION AND FUTURE WORK

This paper presents a reproducible computational framework for fault-tolerant, drag-aware swarm control under radiation-driven sensing and communication degradation. Under the modeled Extreme profile, decentralized M-PSO attains 84.0% success versus 3.0% for a static leader and 1.8% after leader re-election; a separate hot-standby campaign remains at 0.6–1.6% across three pre-specified timeouts despite frequent failover activation. The targeted ablation shows that drag has a modest architecture-independent effect but a much larger effect when combined with decentralized topology. The channel, latch-up, and standby studies sharpen the bounded conclusion: among the leader-based architectures evaluated here, neither transceiver hardening nor one replicated backup resolves the vulnerability induced by a privileged coordination point.

Limitations and future work: The model is 2D and uses a one-seventh-power-law flow approximation, homogeneous agents, and no thermal-hydraulic coupling or hardware validation. N , r_{comm} , β , and Λ_{fail} are swept, but (w, c_1, c_2) remain illustrative; one hot standby does not exhaust multi-backup or quorum-based centralized protocols. Future work includes matched MARL experiments, multi-standby/majority consensus, 3D CFD, relay heterogeneity, and flow-loop validation.

CODE AND DATA AVAILABILITY

The simulator, campaign scripts, seed policy, tests, and result summaries are released with this manuscript at <https://doi.org/10.5281/zenodo.22166217>. Every campaign uses collision-proof SHA-256 condition-identity, per-trial seeding. The release documents the corrected Canonical-PSO drag ablation and includes the seven-cell ablation, re-election, communication-fragility, and hot-standby campaigns.

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